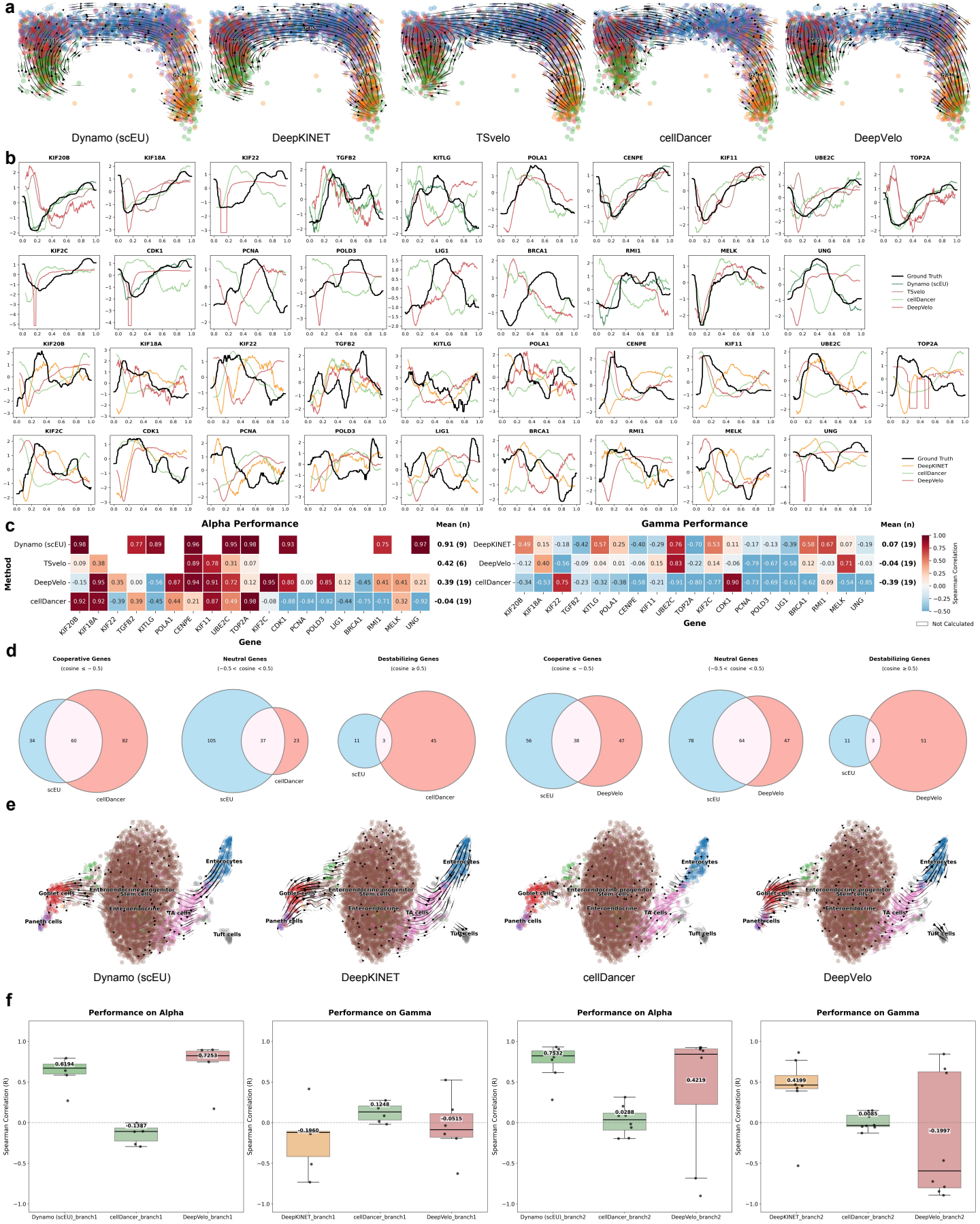

a
Dynamo (scEU)
DeepKINET
TSvelo
cellDancer
DeepVelo
b
KIF20B
KIF18A
KIF22
TGFB2
KITLG
POLA1
CENPE
KIF11
UBE2C
TOP2A
KIF2C
CDK1
PCNA
POLD3
LIG1
BRCA1
RMI1
MELK
UNG
Ground Truth
Dynamo (scEU)
TSvelo
cellDancer
DeepVelo
c
Alpha Performance
Gamma Performance
Mean (n)
Method
Gene
Spearman Correlation
Not Calculated
Dynamo (scEU)
TSvelo
DeepVelo
cellDancer
DeepKINET
0.91 (9)
0.42 (6)
0.39 (19)
-0.04 (19)
0.07 (19)
-0.04 (19)
-0.39 (19)
d
Cooperative Genes
(cosine ≤ −0.5)
Neutral Genes
(−0.5< cosine <0.5)
Destabilizing Genes
(cosine ≥0.5)
scEU
cellDancer
DeepVelo
34
60
82
105
37
23
11
3
45
56
38
47
78
64
47
11
3
51
e
Enterocytes
Goblet cells
Enteroendocrine progenitor
Stem cells
Enteroendocrine
TA cells
Paneth cells
Tuft cells
Dynamo (scEU)
DeepKINET
cellDancer
DeepVelo
f
Performance on Alpha
Performance on Gamma
Spearman Correlation (R)
0.6194
-0.1387
0.7253
-0.1960
0.1248
-0.0515
0.7532
0.0288
0.4219
0.4199
0.0085
-0.1997
Dynamo (scEU)_branch1
cellDancer_branch1
DeepVelo_branch1
DeepKINET_branch1
cellDancer_branch1
DeepVelo_branch1
Dynamo (scEU)_branch2
cellDancer_branch2
DeepVelo_branch2
DeepKINET_branch2
cellDancer_branch2
DeepVelo_branch2